

R Programming Tutorial (Part 4)

Prepared by: Chan Weng Howe

Outline

- Chi-square test using R
 - One-way contingency table
 - Categories with equal frequencies
 - Categories with unequal frequencies
 - Two-way contingency table
 - Chi-square test of independence

- One variable: No. of Accidents
- Categories: 5 (Mon-Fri)
- Significance level: 0.05

$H_0: P_1=P_2=P_3=P_4=P_5$
 H_1 : At least 1 of the 5 proportions is different from others

```
chisq.test(observed_frequency,  
           correct=FALSE)
```

observed data

Chi-squared test for given probabilities

χ^2 statistics of the
observed data

P-value of the χ^2 statistics

One-way Contingency Table (equal probabilities)

Function in R:

```
chisq.test(observed_frequency,
           correction=FALSE)
```

Save result into

output

```
> str(output)
List of 9
 $ statistic: Named num 10.7
 .. attr(*, "names")= chr "X-squared"
 $ parameter: Named num 4
 .. attr(*, "names")= chr "df"
 $ p.value:   num 0.0308
 $ method:   chr "Chi-squared test for given probabilities"
 $ data.name: chr "noAcc"
 $ observed:  num [1:5] 31 42 18 25 31
 $ expected:  num [1:5] 29.4 29.4 29.4 29.4 29.4
 $ residuals: num [1:5] 0.295 2.324 -2.102 -0.811 0.295
 $ stdres:   num [1:5] 0.33 2.598 -2.351 -0.907 0.33
 - attr(*, "class")= chr "htest"
```

str() function use to inspect the structure of a variable in R

```
> output$statistic
X-squared
10.65306

> output$p.value
[1] 0.03075267

> output$parameter
df
4

> output$observed
[1] 31 42 18 25 31

> output$expected
[1] 29.4 29.4 29.4 29.4 29.4
```

Variable in R could be a "List", to access the specific information from the list can use the \$ sign follow by the information name

observed data

```
1 noAcc <- c(31, 42, 18, 25, 31)
2
3 output <- chisq.test(noAcc,
4
5 correct=FALSE)
```

```
6 Chi-squared test for given probabilities
7
8 data: noAcc
9 X-squared = 10.653, df = 4, p-value = 0.03075
```

X² statistics of the
observed data

P-value of the X²
statistics

One-way Contingency Table (equal probabilities)

$$\chi^2 = \sum \frac{(o_i - e_i)^2}{e_i}$$

WHAT WE KNOW?

- One variable: No. of Accidents
- Categories: 5 (Mon-Fri)
- Significance level: 0.05

WHAT WE WANT TO FIND?

$H_0: P_1 = P_2 = P_3 = P_4 = P_5$

H_1 : At least 1 of the 5 proportions is different from others

Function in R:

qchisq(alpha, df, lower.tail=FALSE)

- To get the x2 statistics for the corresponding alpha values/p-values.

pchisq(x2, df, lower.tail=FALSE)

- To get the p-value for the corresponding x2 statistics values.

Goodness-of-Fit test in chi-square always right-tail(upper-tail), so need to set the parameter "lower.tail=FALSE"

```

1  noAcc <- c(31, 42, 18, 25, 31)
2  expprob <- sum(noAcc)/5
3  expAcc <- c(expprob, expprob,
4             expprob, expprob, expprob)
5
6  #calculate x2 for each category
7  exp <- ((noAcc-expAcc)^2)/expAcc
8
9  #sum up to a x2 statistics
10     x2 <- sum(exp)
11
12     #critical value
13     alpha <- 0.05
14     x2.alpha <- qchisq(alpha,
15                        df=4,
16                        lower.tail=FALSE)
17
18     #value of x2 of data
19     pvalue <- pchisq(x2,
20                      df=4,
21                      lower.tail=FALSE)
22

```

One-way Contingency Table (unequal probabilities)

$$\chi^2 = \sum \frac{(o_i - e_i)^2}{e_i}$$

WHAT WE KNOW?

- One variable: Frequencies of M&Ms
- Categories: 6 colors
- Significance level: 0.05
- Claimed p:

$p_1 = 0.30, p_2 = 0.20, p_3 = 0.20, p_4 = 0.10, p_5 = 0.10, p_6 = 0.10$

WHAT WE WANT TO FIND?

$H_0: p_1 = 0.30, p_2 = 0.20, p_3 = 0.20, p_4 = 0.10, p_5 = 0.10, p_6 = 0.10$

H_1 : At least 1 proportions is different from claimed

Function in R:

`chisq.test(observation, p=prob, correct=FALSE)`

In R, `chisq.test()` by default applies "Yates" correction on the chi-square test. Which is arguably unnecessary, so have to set **correct=FALSE**

Since the claim has unequal probabilities, set the probability into the parameter **p** of the function

```

1  mnm <- c(33, 26, 21, 8, 7, 5)
2  prob <- c(0.3, 0.2, 0.2, 0.1,
3      0.1, 0.1)
4
5  # perform chi-square test on the
6  data with their correspond
7  probabilities
8  chisq.test(mnm, p=prob,
9      correct=FALSE)
10
11  #critical value > x2.alpha
12  alpha <- 0.05 [1] 11.0705
13  x2.alpha <- qchisq(alpha,
14      df=5,
15      lower.tail=FALSE)
16
17  > chisq.test(mnm, p=prob, correct=FALSE)
18
19  Chi-squared test for given probabilities
20
21  data: mnm
22  X-squared = 5.95, df = 5, p-value = 0.3111
  
```

22

Two-way Contingency Table (Test of Independence)

WHAT WE KNOW?

- Two variables:
 - Gender: Male / Female
 - Type of Crime: 4 categories
- Significance level: 0.05

WHAT WE WANT TO FIND?

H_0 : Crime type is independent of gender

H_1 : Crime type is not independent of gender

```
1  male <- c(117,150,109,124)
2  female <- c(66,160,168,106)
3  d <- data.frame(male, female)
4
5  # perform chi-square test on the
6  data table
7  chisq.test(d, correct=FALSE)
8
9  #critical value > x2.alpha
10 alpha <- 0.05 [1] 7.814728
11 x2.alpha <- qchisq(alpha,
12 df=3,
13 lower.tail=FALSE)
14
15 > chisq.test(d,correct=FALSE)
16
17      Pearson's Chi-squared test
18
19 data:  d
20 X-squared = 28.511, df = 3, p-value = 2.837e-06
21
22
```

Two-way Contingency Table (Test of Independence)

Example 2

In the built-in data set `survey`, the **Smoke** column records the students smoking habit, while the **Exer** column records their exercise level.

The allowed values in Smoke are "Heavy", "Regul" (regularly), "Occas" (occasionally) and "Never". As for Exer, they are "Freq" (frequently), "Some" and "None".

Test the hypothesis whether the students smoking habit is independent of their exercise level at .05 significance level.

WHAT WE WANT TO FIND?

H_0 : Smoking habit is independent of the exercise level

H_1 : Smoking habit is not independent of the exercise level

Function in R:

`table(list1, list2,...)`

- Group lists into a contingency table

```

1  library(MASS)
2  # get the contingency table
3  tbl = table(survey$Smoke,
4             survey$Exer)
5
6
7  # perform chi-square test on the
8  data table
9  chisq.test(tbl, correct=FALSE)
10
11      #critical val > x2.alpha
12      alpha <- 0.05[1] 7.814728
13      x2.alpha <- qchisq(alpha,
14                        df=3,
15                        lower.tail=FALSE)
16
17      > chisq.test(tbl,correct=FALSE)
18
19      Pearson's Chi-squared test
20
21      data:  tbl
22      X-squared = 5.4885, df = 6, p-value = 0.4828
23
24      Warning message:
25      In chisq.test(tbl, correct = FALSE) :
26      Chi-squared approximation may be incorrect
  
```


Two-way Contingency Table (Test of Independence)

Example 2

In the built-in data set `survey`, the **Smoke** column records the students smoking habit, while the **Exer** column records their exercise level.

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Test the hypothesis whether the students smoking habit is independent of their exercise level at .05 significance level.

WHAT WE WANT TO FIND?

H_0 : Smoking habit is independent of the exercise level

H_1 : Smoking habit is not independent of the exercise level

```
> tbl
```

	Freq	None	Some
Heavy	7	1	3
Never	87	18	84
Occas	12	3	4
Regul	9	1	7

Warning from the chisquare could be due to the small cell values in the contingency table

```
1 library(MASS)
2 # get the contingency table
3 tbl = table(survey$Smoke,
4             survey$Exer)
5
6
7 # perform chi-square test on the
8 data table
9 chisq.test(tbl, correct=FALSE)
10
11 #critical val > x2.alpha
12 alpha <- 0.05[1] 7.814728
13 x2.alpha <- qchisq(alpha,
14                     df=3,
15                     lower.tail=FALSE)
```

```
> chisq.test(tbl,correct=FALSE)
```

Pearson's Chi-squared test

data: tbl

X-squared = 5.4885, df = 6, p-value = 0.4828

Warning message:

In chisq.test(tbl, correct = FALSE) :

Chi-squared approximation may be incorrect

Two-way Contingency Table (Test of Independence)

Example 2

In the built-in data set `survey`, the **Smoke** column records the students smoking habit, while the **Exer** column records their exercise level.

The allowed values in Smoke are "Heavy", "Regul" (regularly), "Occas" (occasionally) and "Never". As for Exer, they are "Freq" (frequently), "Some" and "None".

Test the hypothesis whether the students smoking habit is independent of their exercise level at .05 significance level.

WHAT WE WANT TO FIND?

H_0 : Smoking habit is independent of the exercise level

H_1 : Smoking habit is not independent of the exercise level

```
> tbl2 = cbind(tbl[, 'Freq'], tbl[, 'None'] + tbl[, 'Some'])
> tbl2
```

	[,1]	[,2]
Heavy	7	4
Never	87	102
Occas	12	7
Regul	9	8

Create another table, by merging the 2nd and 3rd columns into single columns

```
1 library(MASS)
2 # get the contingency table
3 tbl = table(survey$Smoke,
4             survey$Exer)
5 # create a new table
6 tbl2 = cbind(tbl[, 'Freq'],
7              tbl[, 'None'] + tbl[, 'Some'])
8
9 # perform chi-square test on the
10 data table
11 chisq.test(tbl2, correct=FALSE)
12
13 #critical val > x2.alpha
14 alpha <- 0.05[1] 7.814728
15 x2.alpha <- qchisq(alpha,
16                    df=3,
17                    lower.tail=FALSE)
```

```
19 > chisq.test(tbl2, correct=FALSE)
```

```
20      Pearson's Chi-squared test
```

```
21 data:  tbl2
```

```
22 X-squared = 3.2328, df = 3, p-value = 0.3571
```

THE END...